MIL-B-21465(NOrd)

15 July 1958

MILITARY SPECIFICATION

BUTYL STEARATE - NORMAL

1. SCOPE

1.1 This specification covers one grade of n-butyl stearate for use in propellant powders.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and publications, of the issue in effect on date of invitation for bids, form a part of this specification.

STANDARDS

Military

MIL-STD-129 -- Marking for shipment and storage
MIL-STD-105 -- Sampling Procedures and Tables for Inspection by Attributes

PUBLICATIONS

Bureau of Ordnance

OP-400 -- General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment (Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications.--The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Society for Testing Materials
ASTM D92-52 Standard Method of Test for Flash and Fire Points by means of Cleveland Open Cup.

FSC 6810

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ASTM D460-54 Standard Methods of Sampling and Chemical Analysis of Soaps and Soap Products.

3. REQUIREMENTS

3.1 Acidity (or stearic acid).--Maximum, 0.5 percent.
3.2 Color.--The color shall be not darker than that of the 300 P.P.M. platinum cobalt standard when determined in accordance with test method in 4.4.2.
3.3 Saponification number.--165-176.
3.4 Iodine number.--Maximum, 6.0.
3.5 Specific gravity at 20°/20°C.--0.855 ± 0.005.
3.6 Titer test.--20.5°C ± 2.0°C.
3.7 Flash point test.--Minimum, 370°F.
3.8 Odor.--Faint, fatty, free of butanol.
3.9 Clarity.--Free of suspended matter.
3.10 Moisture.--No turbidity when 1 volume is mixed with 19 volumes of 66° gasoline.
3.11 Workmanship.--The workmanship shall be in accordance with such practices as to insure compliance with all the requirements of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Lot.--A lot shall consist of material from the same batch or blending operation and subjected to the same processing operation and conditions, from one manufacturer.
4.2 A sample is defined as 1 pound of n-butyl stearate. Sampling shall be performed in accordance with MIL-STD 105, appendix inspection level L-3 with a minimum requirement of five samples per lot, the unit for the determination of sample size being the number of pounds comprising the lot, however no more than 1 sample shall be selected from 1 shipping container unless the sample size (or number of samples) is greater than the number of shipping containers. Each sample shall be identified as to the container, from which it was taken, manufacturer, plant, contract or order number, lot number, number of poinds and number of containers in the lot.
4.3 Inspection.--Inspection shall be made at the point of delivery unless otherwise specified in the contract or order.
4.31 Government Inspection.--The government inspector will make such inspections as are necessary to determine that all materials and services are in accordance with the requirements of the contract, pertinent drawings and specifications. Unless otherwise specified, these inspections will be conducted in

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accordance with instructions contained in OP 400.

4.3.2 Contractor's Inspection.--Contractor's inspection shall be conducted in accordance with provisions of "Inspection by Contractor" as contained in OP 400.

4.4 Tests.--The laboratory tests shall be made in accordance with the following paragraphs. For Navy purchases, the test shall be made at a government laboratory, unless otherwise specified in the contract or order.

4.4.1 Acidity.--A 5 gm. portion of the sample shall be accurately weighed and transferred to a 250 ml. Erlenmeyer flask. In another flask 100 ml. or 95% ethyl-alcohol shall be titrated to a faint pink color with 0.1 N sodium hydroxide solution using phenolphthalein solution as the indicator. A 50 ml. portion of the neutralized alcohol shall be transferred to the flask containing the sample, mixed thoroughly, and titrated rapidly with 0.1 N sodium hydroxide solution to a pink color matching that of the remaining portion of neutralized alcohol. The acid in the sample shall be calculated to percentage stearic acid as follows:

$$\text{Percent stearic acid} \quad \frac{28.4\ VN}{W}$$

where

V = ml. of sodium hydroxide solution required for titration
N = normality of sodium hydroxide solution
W = weight of sample in grams

4.4.2 Determination of color.--

4.4.2.1 Preparation of color standard.--The stock color standard, containing 500-ppm of platinic cobalt chloride (APHA color standard), may be purchased (see 6.2), or may be prepared as follows: A 1.245 gm. portion of potassium chloroplatinate (K_2PtCl_6 containing 0.500 gm. platinum) and 1.000 gm. of crystallized cobaltous chloride ($CoCl_2 \cdot 6H_2O$, containing 0.248 gm. of cobalt) shall be dissolved in distilled water containing 100 ml. reagent grade

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hydrochloric acid. The solution shall be diluted to 1 liter with distilled water, and mixed well. This solution shall have a color intensity equivalent to 500-ppm of platinic cobalt chloride.

4.4.2.1.1 To prepare standards having a desired color, the ml. of 500-ppm solution required for a given color standard shall be calculated by means of the following equation. This volume of standard solution shall be diluted with distilled water to the mark in a Nessler tube, and mixed well.

$$\text{Volume of stock solution required} \quad \frac{AV}{500}$$

where

A = concentration in ppm desired
V = final volume of diluted solution

4.4.2.2 Determination of color.--By means of a pipet, 6 ml. of the 500-ppm standard shall be transferred to a 100 ml. Nessler tube, and diluted to the mark with distilled water. A second Nessler tube shall be filled to the mark with the sample as received. The intensity of the color of the sample shall be compared with the standard by looking vertically downward through the two solutions against a white background. The sample color shall not be darker than the standard.

4.4.3 Saponification number.--A 2 gm. portion of the sample sample shall be accurately weighed and transferred by means of a weighing pipet to a 250 ml. Erlenmeyer flask having a standard taper ground glass neck. Exactly 25 ml. of 0.5 N alcoholic potassium hydroxide shall be added. The flask shall be attached to a reflux condenser by means of a ground glass joint, and the contents boiled gently for approximately one hour. The condenser

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and the ground glass joints shall be washed with approximately 25 ml. of distilled water. The solution shall be cooled to room temperature, 10 drops of phenolphthalein indicator solution added, and the excess potassium hydroxide titrated with 0.3 N hydrochloric acid. A blank shall be run concurrently with the sample. The saponification number for butyl stearate shall be calculated as follows: Saponification number

$$\frac{56.1 \times (A-B)\ N}{W}$$

where

A = ml. of acid used to titrate the blank
B = ml. of acid used to titrate the sample
N = normality of acid
W = weight of sample

4.4.4 Iodine number.--

4.4.4.1 Acetic acid for Wijs solution.--The glacial acetic acid shall be free from reducing impurities, having a melting point of $14.8^{\circ} \pm 0.1^{\circ}C$ and contain 99 percent acetic acid. It shall be tested and prepared as follows: If these requirements are not compiled with, the results of the iodine number determination will be unreliable.

4.4.4.1.1.1 Test for reducing impurities (potassium permanganate test.--A 2 ml. portion of the acetic acid shall be diluted with 10 ml. of distilled water, 0.1 ml. of 0.1 N potassium permanganate solution added and a temperature of $24^{\circ} \pm 3^{\circ}C$ maintained. The acid shall be considered satisfactory with respect to reducing impurities for use in this determination if the pink color is not completely discharged at the end of 2 hours.

4.4.4.1.1.2 Melting point determination.--A 6 inch test tube shall be filled about two-thirds full with acetic acid. A standardized thermometer equipped with a cork stopper, shall be inserted into the acid in such a manner that the cork stopper fits into the test tube. The amount of acid shall be at least double the quantity required to cover the bulb of the thermometer when the bottom of the latter is

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one-half inch from the bottom of the test tube. The test tube shall be suspended within a larger test tube by means of a cork. The acid shall be cooled by immersing the assembly in ice water until the temperature is 10°C. The assembly is them withdrawn from the ice bath and the acid stirred vigorously for a few minutes, thus causing the supercooled liquid to crystallize partially and give a mixture of liquid and acid crystals. Thermometer readings shall be taken every 15 seconds and that temperature at which the reading remains constant for 2 minutes shall be considered the true melting point. In case the melting point is found to be less then 14.8°C the acid shall not be used. Should the melting point be greater than 14.8°C, small quantities of distilled water shall be added to reduce the melting point to 14.8°C. The melting point shall be determined after each addition of water, thus making sure that the final melting point of the acid is 14.8°C ± 0.1°C.

4.4.4.1.2 Chloroform.--U.S.P. grade, acetone-free, chloroform.

4.4.4.1.3 Wijs solution.--The preparation of the iodine monochloride solution presents no great difficulty, but it should be done with care and accuracy in order to obtain satisfactory results. There shall be in the solution no sensible excess either of iodine or more particularly, of chlorine over that required to form the monochloride. This condition is most satisfactorily attained by dissolving in the whole the acetic acid to be used in the requisite quantity of iodine, using gentle heat to assist the solution if it is found necessary. Iodine shall be dissolved in glacial acetic acid in the proportion so that 18 gm. of iodine shall be present in 1000 ml. of solution. A small portion of this solution shall be set aside while pure, and dry chlorine passed into the remainder until the halogen content of the solution is doubled. Ordinarily it will be found that by passing the chlorine into the main part of the solution until the characteristic color of free iodine has just been discharged, there will be a

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slight excess of chlorine which will be corrected by the addition of the requisite amount of the unchlorinated portion until all free chlorine has been destroyed. A slight excess of iodine does little or no harm, but excess of chlorine must be avoided. As an example, if the titration of 20 ml. of the original iodine-acetic acid solution required 22 ml. of 0.1 N thiosulfate, 20 ml. of the finished Wijs solution should require between 43.5 and 44 ml. of the thiosulfate solution.

4.4.4.1.4 Starch solution.--A mixture of 2 gm. of the soluble starch, several mg. of mercuric iodide preservative and a little water shall be prepared. The mixture shall be added slowly to 500 ml. of boiling water, boil for 5 minutes and then cooled.

4.4.4.1.5 Standard sodium thiosulfate solution.--Reagent-grade sodium thiosulfate shall be dissolved in distilled water, that has been well boiled to free it from carbon dioxide, in the proportion of 24.84 gm. of crystallized sodium thiosulfate to 1000 ml. of solution. 0.1 gm. of sodium carbonate shall be added in the solution and allow to stand overnight before standardized by any suitable standard method. The reagent shall be preserved in a stock bottle provided with a guard tube filled with soda lime.

4.4.4.1.6 Potassium iodide solution.-- Potassium iodide, free from iodate, shall be dissolved in distilled water in the proportion of 150 gm. of potassium iodide to 1000 ml. of the solution.

4.4.4.2 Procedure.--An accurately weighed portion of 0.12 to 0.16 gm. of the sample shall be transferred to a 250 ml. bottle having a well-ground glass stopper, or an Erlenmeyer flask having a specially flanged neck for the iodine test. Ten ml. portion of chloroform shall be added to the flask and the flask whirled to dissolve the sample. A 10 ml. portion of chloroform shall be added to each of two empty flasks like that used for the sample. By means of a pipet, 20 ml. of the Wijs solution shall be added to each flask and the latter allowed to stand with occasional

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shakings for 60 ± 5 minutes in a dark place at a temperature of from 21° to 23°C. At the end of this period, 10 ml. of water shall be added to the flask and the solution titrated with standard sodium thiosulfate, using starch as the indicator. The titration on the two blank tests shall agree within 0.1 ml. From the differences between the average of the blank titrations and the titration of the sample, the iodine number of the material shall be calculated as follows:

$$\text{Iodine number} \quad \frac{12.7\ N\ (V-v)}{W}$$

where

N = normality of sodium thiosulfate solution used to titrate the blank
V = ml. of sodium thiosulfate used to titrate the blank
v = ml. of sodium thiosulfate used to back titrate the sample solution
W = gm. sample

4.4.5 Specific gravity.--The specific gravity shall be determined at 20°C by means of pycnometer or hydrometer.

4.4.6 Titer test.--

4.4.6.1 Apparatus.--The apparatus required is shown in ASTM-D 460-54 and consists of the following:

4.4.6.1.1 Water bath.--A 2 liter Griffin low form beaker.

4.4.6.1.2 Wide-mouth bottle.--A wide-mouth 450 ml. bottle, 190 mm. in height, having a neck 38 mm. in inside diameter, fitted with a cork to carry the test tube and with sufficient weights or lead in the bottom of the bottle to hold it steady when placed in the water bath.

4.4.6.1.3 Test tube.--Test tube 100 mm. in length, and 25 mm. in diameter with or without rim. The tube may have an etched line extending around it at a distance of 57 mm. from the bottom to indicate the height to which it is to be filled.

4.4.6.1.4 Stirrer.--A stirrer 2 to 3 mm. in outside diameter and with one end bent in the form of a loop 19 mm. in diameter; glass, nichrome, stainless steel, or monel metal wire may be used. The upper end of the stirrer may be formed to accomodate hand stirring or for attachment to a mechanical stirrer.

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4.4.6.1.5 Laboratory thermometer.--A laboratory thermometer having a range of 0 to 150°C.

4.4.6.1.6 Titer test thermometer.--An A.S.T.M.: F.A.C. Titer test thermometer, graduated in Centigrade degrees, having a range of -2 to +68°C. and conforming to the requirements for thermometer 36C as prescribed in the Standard Specifications for A.S.T.M. Thermometers (A.S.T.M. Designation: E 1).

4.4.6.2 Procedure.--The test tube shall be filled to the 57 mm. line with sample and placed in the wide-mouth bottle. The bottle assembly shall be placed in the water bath (the water should reach the neck of the bottle), and the temperature of the bath adjusted to 5°C. The titer test thermometer is inserted in the center of the sample and suspended at such a height that the immersion mark, coincides with the top of the sample. The sample shall be stirred with the stirring rod in a vertical manner at the rate of 100 completed up-and-down motions per minute. The stirrer should travel about 3.8 cm. The thermometer shall be read every 15 seconds and the stirring continued until the temperature remains constant (or begins to rise) for 30 seconds. Stirring shall be discontinued immediately and the rise in temperature reached by the thermometer shall be reported as the titer.
Duplicate determinations shall not differ by more than 0.2°C.

4.4.7 Flash point test.--

4.4.7.1 Apparatus.--The apparatus shall consist of the following:

4.4.7.1.1 Cleveland flash cup.--A Cleveland open cup is made of brass and conforms to the dimensional requirements prescribed in Table I of ASTM-D92-52. The leveled edge of the cup shall be at an angle of approximately 45 degrees. There may be a fillet of approximately 5/32 inch (0.397 cm.) in radius inside the bottom of the cup.

4.4.7.1.2 Heating plate.--A metal plate, as shown in Figure 3 of ASTM-D92-52, 1/4 inch (0.635 cm.) in thickness, and 6 inches (15.24 cm.) in width for supporting the flask cup. The plate shall be of brass, cast iron, wrought iron, or steel. In the center of the plate there shall be a plane depression 1/32 inch (0.079 cm.) in depth, and of just sufficient diameter to fit the cup. There shall be cut in the center of the asbestos board a circular hole just fitting the cup. Heat may be supplied from any convenient

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source. The use of a gas burner, electric heater, or alcohol lamp is permitted, but under no circumstances are products of combusion of free flame allowed to come up around the cup. The source of heat shall be centered under the opening in the plate and shall be of a type that will not produce local super-heating. If a flame heater is used, it may be protected from drafts or excessive radiation by any suitable type of shield that does not project above the level of the upper surface of the asbestos board.

4.4.7.1.3 Thermometer.--An ASTM Open Flash Thermometer, graduated in either Centigrade or Fahrenheit degrees as specified, having a range of -6 to +400°C or +20 to 760°F and conforming to the requirements for thermometer LLC and LLF, respectively, as prescribed in the Specifications for ASTM Thermometers (ASTM Designation: E 1).

Note.--Whenever thermometers complying with ASTM requirements are not available, thermometers complying with their requirements for the Institute of Petroleum Thermometer "IP 28F (or 28C) Cleveland" may be used, provided claibration corrections are used in temperature ranges where the Ip requirements for scale accuracy are less stringent than those of ASTM.

4.4.7.2 Procedure.--The thermometer shall be suspended in a vertical position by any suitable device. The bottom of the bath shall be 1/4 inch (0.635 cm.) from the bottom of the cup, and above a point halfway between the center and back of the cup.

The test flame shall be approximately 5/32 inch (0.397 cm.) in diameter.

Note.--For purpose of comparison it is recommended that a bead of suitable light-colored material be mounted in a convenient position so that the size of the test flame can be determined. The device for applying the flame may be of any suitable type but it is suggested that the tip be approximately 1/16 inch (0.159 cm.) in diameter at the end and that the orifice be 1/32 inch (0.079 cm.) in diameter. If the device for operating the test flame be mounted in such a manner as to permit automatic duplication of the sweep of the test flame the radius of swing shall be not less than 6 inches.

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The test flame shall be applied as the temperature read on the thermometer reaches each successive 5F (2.8 C) mark. The flame shall be passed in a straight line (or on the circumference of a circle having a radius of at least 6 inches) across the center of the cup and at right angles to the diameter passing through the thermometer. The test flame shall, while passing through the surface of the sample, be in the plane of the upper edge of the cup. The time for the passage of the test flame across the cup shall be approximately 1 sec. The sample shall be heated at a rate not exceeding 30F (16.7 C) per minute temperature rise, until a point is reached approximately 100 F (55.6 C) below the probable flask point of heating the sample. Thereafter, the rate of heating shall be decreased and for at least the last 50 F (27.8 C) before the flash point is reached, the rate shall be not less than 9 F (5 C) nor more than 11 F (6.1 C) per minute. Flash point.--Take the flash point as the temperature read on the thermometer when a flash appears at any point on the surface of the sample. The true flash must not be confused with a bluish halo that sometimes surrounds the test flame.

4.4.7.3 Precautions.--Laboratory conditions shall be such that flash point tests are made in a room or compartment free from air currents. Care shall be taken while heating to avoid disturbing the vapors evolved in the cup, either by careless breathing or unnecessary movements near the flash point apparatus.

4.5 Lot acceptance.--The lot shall be accepted provided that all the samples satisfy the requirements of paragraph 3, when tested in accordance with paragraph 4.4. A lot shall be rejected if one or more samples fail to satisfy these requirements.

4.6 Resubmission and retest.--If a lot is rejected, the contractor has the option of having a retest performed on the lot, at no expense to the government. Before resubmitting the lot for retest, the contractor shall remove all defective portions of the lost and present it as a resubmitted lot. The procedures of paragraph 4.2 and 4.5 apply. If a resubmitted lot is rejected on retest, it may not be resubmitted again.

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5. PREPARATION FOR DELIVERY

5.1 Packing, Level C (domestic shipment, items for immediate use) unless otherwise specified in the contract or order, N-butyl stearate shall be packed in standard steel or aluminum containers, so constructed as to afford protection against damage during shipment, from the supply source to the first receiving activity for immediate use. The containers shall comply with Consolidated Freight Classification Rules or other common carrier's regulations, and be so constructed to insure acceptance by common or other carriers at the lowest freight rate.

5.2 Marking.--In addition to any special marking required by the contract or order, each container shall be marked in accordance with STANDARD MIL-STD-129, and in accordance with the follwoing information: Material, Specification No., Quantity, Contractor, Manufacturer, Contract No., Gross Weight, Date, Lot No.

6. NOTES

6.1 Ordering data.--Procurement documents should specify the title, the number and the date of the specification.

6.2 The stock color standard, containing 500 ppm of platinic cobalt chloride (APHA) color standard, a product of Hartman Laddon Co., Philadelphia, Pa., has been found satisfactory for use in the color determination (See 4.4.2.1).

NOTICE.--When government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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